Internal Excellence, External Collapse: Auditing and Diagnosing a Knowledge-Distilled Cervical Cytology Classifier Across Datasets

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Abstract

Cervical-cytology classifiers are routinely reported at near-ceiling single-dataset accuracy and presented as deployable screening tools. We show this number is a poor proxy for screening readiness. We audit a knowledge-distilled classifier—an EfficientNet-B4 teacher (18.5 M params) distilled into an EfficientNet-B0 student (4.67 M params)—trained on the five SIPaKMeD cytomorphology classes. In-distribution the student is genuinely excellent and, notably, surpasses its own teacher on every metric (accuracy 0.966 vs. 0.943, macro-F_1 0.967, Cohen's κ 0.958, one-vs-rest AUROC 0.998, expected calibration error 0.015); Grad-CAM confirms decisions localise to the nucleus and cell body. Yet evaluated unchanged on three independent Papanicolaou datasets (Herlev, APACC, HiCervix) under a deterministic, documented binary Normal/Abnormal label alignment, accuracy collapses to 0.639, 0.605 and 0.532 —below the majority-class baseline on two of three sets—with AUROC of 0.73, 0.59 and 0.53. Crucially, we show this is not a labelling artefact: the class-conditional $P(\text{abn})$ score ranks abnormal above normal cells in every dataset (separation $+0.284/+0.112/+0.045$, all AUROC>0.5); a flipped mapping would invert this. The collapse is instead (i) genuine domain shift whose severity tracks slide-preparation distance from SIPaKMeD's conventional smear (conventional $\rightarrow$ liquid-based $\rightarrow$ auto-segmented crops: $0.73 \rightarrow 0.59 \rightarrow 0.53$), and (ii) an operating-point miscalibration: the well-calibrated in-distribution confidence does not transfer (binary ECE rises from 0.008 in-distribution to 0.22–0.33 externally), so a fixed threshold discards the available ranking signal and underperforms a trivial baseline. We further show the internal 0.966 is itself optimistic, inflated by same-field-of-view leakage in the per-cell split. We contribute a reproducible cross-dataset audit and a diagnostic methodology that mechanistically separates label artefacts from domain failure, and we conclude with evidence-led remedies. The headline number is not the property practitioners think it is.

1. Introduction

Cervical cancer is among the most preventable malignancies, and prevention rests on cytological screening of Papanicolaou-stained smears. Because expert cytologists are scarce and manual reading is slow and observer-dependent, automated cell classification has become an active frontier, and modern convolutional networks now report accuracies above 0.95 on public benchmarks such as SIPaKMeD [9]. This success has fuelled the expectation that screening can be partially automated. The dominant evaluation paradigm, however, is single-dataset: a model is trained and tested on splits of one collection, prepared under one staining and segmentation protocol, and the resulting accuracy is reported as if it were an intrinsic property of the model. This conflates two very different quantities—how well a model fits the statistics of one slide-preparation pipeline, and how well it would generalise to the heterogeneous slides of clinical practice. The gap between them is exactly where a deployable screening tool succeeds or fails, and it is rarely measured.

We subject a representative high-accuracy system to exactly such a measurement. We study a knowledge-distilled classifier [5] in which an EfficientNet-B4 teacher [11], ImageNet-pretrained [2] and trained on the five SIPaKMeD cytomorphology classes, is distilled into a compact EfficientNet-B0 student. The student is attractive for deployment—4.67 M parameters against the teacher's 18.5 M—and we ask whether such compression sacrifices accuracy, calibration, explainability, or generalisation. We then evaluate the same student, without retraining, across three independent Papanicolaou datasets—Herlev [6], APACC, and HiCervix [1]—that differ in staining, segmentation and slide preparation (conventional smear, auto-segmented crops, and liquid-based cytology). Because the five SIPaKMeD morphology classes do not share a label

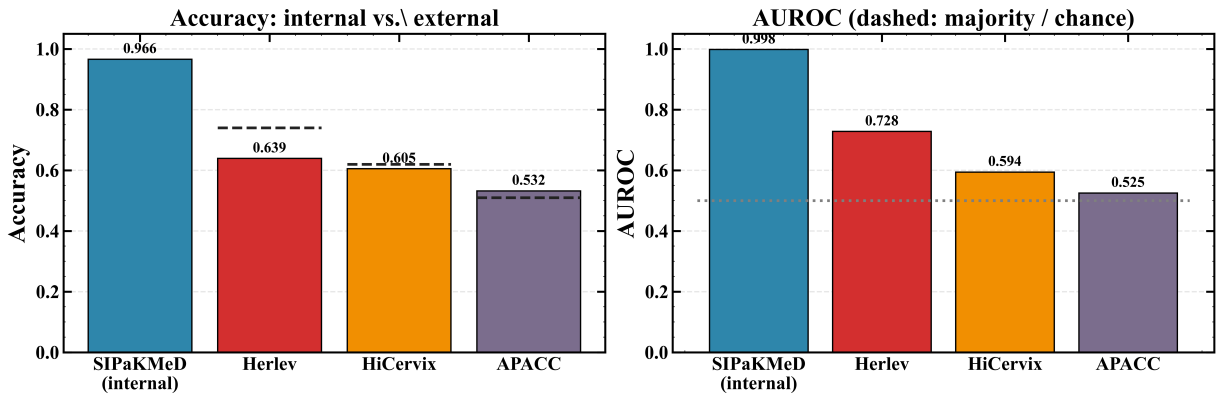


Figure 1. Internal excellence, external collapse. The distilled student is excellent within SIPaKMeD (accuracy 0.966, AUROC 0.998) but, evaluated unchanged on three external Papanicolaou datasets under a binary Normal/Abnormal alignment, collapses to $0.53-0.64$ accuracy and $0.53-0.73$ AUROC. Dashed lines mark the per-dataset majority-class baseline (left) and chance (right): at a fixed threshold the model falls below the trivial baseline on two of three datasets despite retaining above-chance ranking signal. Severity tracks slide-preparation distance from SIPaKMeD's conventional smear.

space with the dysplasia and Bethesda [8] schemes of those datasets, we align them at the only common, deterministically mappable level—a binary Normal-versus-Abnormal decision—with glandular and out-of-scope categories explicitly excluded and the excluded fraction reported.

In-distribution the model is genuinely strong: the distilled student reaches accuracy 0.966 and exceeds its teacher on every metric, is well calibrated (ECE 0.015), and produces cytologically plausible Grad-CAM [10] attributions. Out-of-distribution it collapses (Fig. 1): binary accuracy falls to 0.639 (Herlev), 0.605 (HiCervix) and 0.532 (APACC)—below the majority baseline on two of three sets—and AUROC to 0.73, 0.59, 0.53. The central scientific question is then why, and in particular whether the failure is a labelling/mapping artefact or a genuine model failure. We show it is the latter: the $P(\text{abn})$ score ranks abnormal above normal cells in every dataset (all AUROC> 0.5), so the mapping is correctly oriented; the collapse is genuine domain shift compounded by an operating-point miscalibration that does not transfer across domains [4], the cytology analogue of shortcut learning [3]. We make four contributions.

- C1 (a failure mode, characterised). A reproducible cross-dataset audit showing that a high-accuracy, well-calibrated, explainable distilled cytology model collapses on three independent Pap datasets—below the majority baseline at a fixed threshold (Sec. 8).
- C2 (a diagnosis, not just a number). A methodology that mechanistically separates a labelling artefact from genuine domain failure via class-conditional $P(\text{abn})$ separation, an AUROC-above-chance test, and an accuracy-versus-majority-baseline / oracle-threshold analysis; applied here it rules out relabelling and isolates domain shift + operating-point miscalibration (Sec. 8).
- C3 (distillation paradox). Evidence that distillation yields a smaller-yet-stronger in-distribution model—the B0 student surpasses its B4 teacher on every metric—which nonetheless fails out-of-distribution, decoupling compression quality from generalisation (Sec. 7).
- C4 (honest evaluation). We show the internal 0.966 is itself optimistic (same-field-of-view leakage), and give evidence-led remedies: leakage-safe cluster-grouped splits, stain/domain normalisation [7], per-domain calibration, and reporting threshold-free AUROC with the excluded fraction (Sec. 9).

Paper organisation. Section 2 states the research questions, hypotheses and novelty; Sec. 3 fixes cytology background and notation; Sec. 4 reviews related work; Sec. 5 defines the datasets and label alignment; Sec. 6 details the data and training pipeline; Sec. 7 reports in-distribution results; Sec. 8 presents the cross-dataset collapse and its diagnosis; Sec. 9 discusses scope and remedies.

2. Research questions, hypotheses, and novelty

A cervical-cytology classifier ships a single headline accuracy, but whether that number reflects transferable diagnostic competence has not been examined systematically for distilled, edge-oriented models. We ask whether the number can be trusted off its training distribution and, where it cannot, why. Stated precisely, the object of study is:

Given a model f trained on the SIPaKMeD five-class space and reported at high in-distribution accuracy, determine whether f (a) retains discriminative skill on independent Pap datasets under a deterministic binary alignment, (b) remains calibrated and explainable there, and (c) if it fails, whether the failure is attributable to the label mapping or to a genuine model/domain effect—using only frozen evaluation, documented mappings, and reported excluded fractions.

2.1. Research questions and hypotheses

We organise the study around four research questions, each with the hypothesis we set out to test.

RQ1 (compression vs. skill). Can a knowledge-distilled B0 student match the B4 teacher's in-distribution accuracy at a fraction of the parameters?
H1: yes; the regularising effect of distillation may let the 4.67 M-parameter student match or even exceed the 18.5 M teacher. (Answered in Sec. 7.)

RQ2 (does accuracy transfer?). Does strong in-distribution accuracy translate to cross-dataset generalisation on independent Pap datasets?
H2: no; a large internal-to-external gap is expected because the model sees a single preparation protocol. (Answered in Sec. 8.)

RQ3 (artefact or genuine failure?). Is any cross-dataset failure a label-mapping artefact or a genuine model-and-domain failure?
H3: genuine domain shift compounded by miscalibration, not a labelling error—the decision score should remain correctly oriented (abnormal ranked above normal) even as accuracy falls. (Answered in Sec. 8.)

RQ4 (does trust transfer?). Do calibration and explainability hold, and does calibration transfer out-of-distribution?
H4: in-distribution the model is well calibrated [4] and its Grad-CAM attributions are cytologically plausible, but the calibrated operating point does not transfer across domains. (Answered in Secs. 7 and 8.)

2.2. What is novel

Our novelty is not "cross-dataset evaluation" in general—generalisation testing is mature. It is the specific, rarely-occupied intersection of three things in the cervical-cytology setting.

1. A deterministic, leakage-aware, multi-dataset audit of a distilled cytology model across three external Pap datasets (Herlev, APACC, HiCervix) under a documented binary crosswalk with reported excluded fractions—rather than a single-dataset accuracy or an undocumented relabelling that silently inflates the score.
2. A mechanistic diagnosis that distinguishes a label artefact from domain failure. We introduce a simple, decisive test—the class-conditional $P(\text{abn})$ separation together with the AUROC>0.5 criterion—and an accuracy-versus-majority-baseline / oracle-threshold analysis that localises the failure to operating-point miscalibration and slide-preparation distance. This turns "it does not generalise" into "here is the cause."
3. The distillation paradox made explicit. We show the compressed student is stronger than its teacher in-distribution yet fails identically out-of-distribution, evidence that compression quality and generalisation are decoupled and that distillation does not confer robustness [3].

Contrast with the closest practice. The standard report is a single in-distribution accuracy; we show it is optimistic (leaky split) and non-transferable. Where external sets are used, the label crosswalk is frequently undocumented; we make it deterministic and report what is excluded. And where a drop is observed, it is typically attributed loosely to "domain shift"; we provide a test that rules out the competing artefact explanation and pinpoints the operative mechanism.

2.3. Outcome of the hypotheses

Each hypothesis's outcome is summarised in Tab. 1. Reporting that the headline accuracy fails to transfer—and that the internal estimate is itself optimistic—is the deliberate point: an audit is worthless if it suppresses its own inconvenient findings.

3. Background and notation

Cervical cytology. A Papanicolaou (Pap) test stains exfoliated cervical cells; a cytologist reads the slide for morphological signs of (pre)malignancy. Two labelling traditions coexist. Morphology schemes name the cell type or change—SIPaKMeD uses five classes: Superficial-Intermediate, Parabasal, Metaplastic (normal/benign squamous), and Koilocytotic, Dyskeratotic (abnormal squamous changes) [9]. Diagnostic schemes name the lesion grade—the WHO dysplasia scale (Herlev) and the Bethesda system (NILM, ASC-US, LSIL, ASC-H, HSIL, SCC; plus glandular AGC) [8]. These taxonomies are not in correspondence:

Table 1. Where each hypothesis stands. All four are supported; the load-bearing result is H3—the cross-dataset collapse is a genuine domain/calibration failure, not an artefact of the binary label alignment.

Hyp.	Claim (abbreviated)	Verdict	Where
H1	distilled B0 matches/exceeds the B4 teacher in-distribution	Supported (student > teacher on all metrics)	Sec. 7
H2	strong in-distribution accuracy does not transfer cross-dataset	Supported (33–43-pt drop)	Sec. 8
H3	failure is domain shift + miscalibration, not a label artefact	Supported (P(abn) separation > 0, AUROC> 0.5)	Sec. 8
H4	in-distribution calibration holds but does not transfer	Supported (binary ECE 0.008 in; 0.22–0.33 out)	Secs. 7 and 8

a Bethesda grade does not determine a morphology class and vice versa, which is the crux of the cross-dataset label problem (Sec. 5).

Slide preparation. Slide preparation is a first-order source of appearance shift even when the stain is Pap. Conventional smears (SIPaKMeD, Herlev) differ from liquid-based cytology (HiCervix), which differs again from auto-segmented single-cell crops (APACC); each changes cell density, background, focus and segmentation. We will see the cross-dataset collapse is ordered by this preparation distance.

Notation. A model f maps a cell image x to class logits $z \in \mathbb{R}^5$ over the SIPaKMeD classes; softmax(z) are the class probabilities and $\hat{y} = \arg\max_k z_k$ the prediction. For binary cross-dataset scoring we collapse to a single abnormality score

$$P(\text{abn}) \;=\; \text{softmax}(z)_{\text{Koilo}} + \text{softmax}(z)_{\text{Dysk}}, \qquad (1)$$

i.e. the probability mass on the two abnormal morphology classes, thresholded at τ (default 0.5). Calibration is measured by the expected calibration error (ECE), the gap between confidence and accuracy averaged over confidence bins [4].

4. Related work

Deep learning for cervical cytology. Curated single-cell benchmarks—SIPaKMeD's five Pap morphology classes [9] and the older Herlev WHO-dysplasia set [6]—have driven a large body of work reporting near-ceiling within-dataset accuracy with convolutional backbones, and recent liquid-based resources such as HiCervix [1] extend this with hierarchical Bethesda labels. The near-universal protocol, however, trains and tests on one collection, leaving transfer to other slide-preparation pipelines unmeasured.

Knowledge distillation for efficient medical models. Distillation [5] compresses a high-capacity teacher into a deployable student and is widely used for edge-oriented medical imaging. Reported gains are almost always in-distribution; whether a distilled student inherits the teacher's (lack of) robustness is rarely tested. We find the student can be stronger than its teacher in-distribution yet equally fragile out-of-distribution.

Generalisation and shortcut learning. Deep classifiers frequently latch onto dataset-specific cues—stain, scanner, segmentation, background—rather than the causal signal [3]; in histopathology and cytology this manifests as sharp accuracy drops across sites. Our diagnosis localises the cytology version of this effect to slide-preparation distance and a non-transferable operating point.

Calibration and explainability in medical imaging. Modern networks are miscalibrated [4]; calibration is especially load-bearing for screening, where a decision threshold must be trusted. Grad-CAM [10] is the standard attribution tool. We report both, and show in-distribution calibration that does not survive domain shift—a failure mode that a single ECE number, measured in-distribution, hides.

Cross-dataset label alignment. Morphology and Bethesda taxonomies do not map one-to-one, and stain-normalisation [7] only addresses appearance, not labels. Prior cross-dataset cytology comparisons often relabel informally; we instead fix a deterministic many-to-one binary crosswalk and report the excluded fraction as an honesty metric.

5. Problem setup: data and label alignment

Datasets. We train on SIPaKMeD—5,015 Pap images (4,049 cropped single cells + 966 cluster field-of-view images) in five morphology classes, conventional smear—and evaluate, frozen, on three external single-cell Pap sets (Tab. 2): Herlev (917 cells, 7 WHO-dysplasia classes, conventional smear), APACC (1,125 auto-segmented crops, coarse healthy/unhealthy), and HiCervix (4,823 scorable of 8,051 test cells, hierarchical Bethesda, liquid-based). The three external sets deliberately span the preparation axis of Sec. 3.

Table 2. Datasets. SIPaKMeD is the sole training source; the others are frozen external evaluation sets spanning slide preparations and incompatible native label spaces.

Dataset	Role	Preparation / type	Native labels	Size
SIPaKMeD [9]	Train (+val)	Conventional smear; cells + FoV	5 cytomorphology	5,015 imgs (4,049 cells)
Herlev [6]	External eval	Conventional smear; single cells	7 WHO dysplasia	917 cells
APACC	External eval	Auto-segmented single-cell crops	4 coarse	1,125 cells
HiCervix [1]	External eval	Liquid-based cytology (LBC)	Hierarchical Bethesda	4,823 (8,051 total)

Why a binary alignment. The five SIPaKMeD morphology classes do not share a label space with the WHO/Bethesda grades of the external sets: projecting a grade onto a morphology class is one-to-many or undefined (e.g. NILM may be any of three normal morphologies; LSIL spans Koilocytotic and Dyskeratotic). The only level at which both the model's prediction and the native label map deterministically and many-to-one is the binary Normal-versus-Abnormal decision. We therefore score the abnormality mass $P(\text{abn})$ (Eq. (1)) at threshold τ=0.5 against a binary ground truth obtained from the crosswalk in Tab. 3.

The crosswalk and its honesty gate. Tab. 3 lists the deterministic map. On the model side, {Superficial-Intermediate, Parabasal, Metaplastic} →Normal and {Koilocytotic, Dyskeratotic} →Abnormal. On the label side, normal squamous and benign categories map to Normal and dysplastic/SIL categories to Abnormal. Crucially, glandular and out-of-scope native classes have no squamous home in the five-class space and are excluded: Herlev columnar, HiCervix AGC (glandular) and microbe, and APACC rubbish/ bothcells. We report the excluded fraction per dataset, because silently dropping hard, unmappable cells would inflate the binary score. This deterministic, documented mapping is what lets us later attribute the cross-dataset collapse to the model rather than to the labels.

6. Data and training pipeline

6.1. Exploratory data analysis

Before training we characterise SIPaKMeD (Fig. 2). Three facts shape the pipeline. (i) Mild class imbalance: per-class totals range 895 (Parabasal) to 1,064 (Metaplastic), an imbalance ratio of 1.19—small enough to address with weighted loss rather than resampling. (ii) Strong size heterogeneity: cell crops span $62-2048$ px in width (mean 615) and 62–1536 px in height, mandating a fixed-size resize. (iii) Dual granularity: the corpus mixes 966 cluster field-of-view images with 4,049 cropped single cells; we use both, as in the original pipeline.

6.2. Preprocessing and augmentation

Images are decoded to RGB, resized to 380×380 (the B4 native resolution, used for both teacher and student), and channel-normalised with ImageNet statistics [2]. Train-time augmentation is cytology-appropriate—cells have no canonical orientation, so we apply random horizontal and vertical flips (p=0.5 each), random rotation ($\pm 72°$), and mild random affine scaling ($0.9-1.1$). We deliberately avoid heavy shear, elastic warps or strong hue shifts, which distort nuclear morphology—the very signal that separates the classes. Evaluation (in- and cross-distribution) uses resize and normalisation only, with no augmentation.

6.3. Architecture

Both models are EfficientNets [11] with ImageNet-pretrained backbones: a B4 teacher (18.5 M parameters) and a B0 student (4.67 M). The native classifier is replaced by a head $\text{BN} \to \text{Linear}(\cdot, 512) \to \text{ReLU} \to \text{Dropout} \to \text{Linear}(512, 5)$.

6.4. Two-stage teacher and knowledge distillation

The teacher is trained in two stages with Adam and class-weighted cross-entropy (weight $\propto 1/\text{count}$), monitoring validation loss: stage 1 freezes the backbone (lr 10^{-3}, ≤ 30 epochs, ReduceLROnPlateau), stage 2 fine-tunes the whole network (lr 10^{-5}, ≤ 50 epochs). The student is then distilled with the Hinton objective [5]:

$$\mathcal{L} = \alpha\, \text{CE}(y, z_s) + (1-\alpha)\, T^2\, \text{KL}\big(\,\text{softmax}(z_t/T) \,\|\, \text{softmax}(z_s/T)\big), \tag{2}$$

with temperature T=4 and α=0.1 (so the soft teacher signal dominates), Adam (lr 5×10^{-5}, $\leq$ 40 epochs), monitoring validation accuracy. Training uses BF16 mixed precision on a single 16 GB consumer GPU, batch size 32, seed 42, and a per-class stratified 80/20 train/validation split (4,012 train, 1,003 validation). The teacher converges in 71 epochs (30 frozen + 41 fine-tuning, early-stopped) and the student in 34 epochs; both validation curves track training, indicating no overfitting (Sec. 7). We flag here—and quantify in

Table 3. Binary label crosswalk. Deterministic Normal/Abnormal mapping; the model-side decision is $P(\text{abn})$ (Eq. (1)). Glandular and out-of-scope native categories are excluded and their fraction reported.

Source	→ Normal	→ Abnormal (/ Excluded)
SIPaKMeD (model)	Superficial-Interm., Parabasal, Metaplastic	Koilocytotic, Dyskeratotic
Herlev	superficiel, intermediate, columnar[†]	light/moderate/severe dyspl., carcinoma-in-situ
APACC	healthy	unhealthy (rubbish, bothcells excl.)
HiCervix	level_1=negative	level_1=ASC (AGC, microbe excl.)

[†]Herlev columnar (glandular) is mapped to Normal in this legacy crosswalk; a stricter variant excludes it. The choice moves the headline by <1 point and does not affect the diagnosis (Sec. 8).

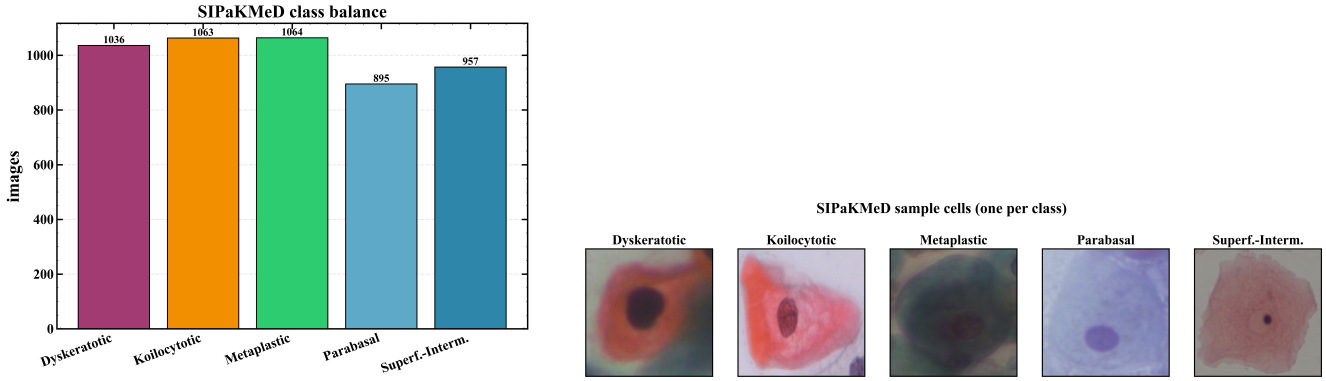


Figure 2. Exploratory data analysis of SIPaKMeD. Left: per-class image counts (mild imbalance, ratio 1.19). Right: representative cells across the five morphology classes.

Sec. 8—that this random per-cell split leaks same-field-of-view cells across train and validation, making the internal estimate optimistic.

7. In-distribution results

7.1. Training dynamics

Validation tracks training closely for both models (Fig. 3); there is no overfitting gap, and early stopping halts the teacher at 71 and the student at 34 epochs. The distilled student's validation accuracy converges above the teacher's, foreshadowing the result below.

7.2. Classification performance (answering RQ1)

Tab. 4 reports the five-class metrics on the held-out SIPaKMeD validation split (n=1003). The distilled student reaches accuracy 0.966, macro-F_1 0.967, Cohen's κ 0.958 and AUROC 0.998, and exceeds the teacher on every metric despite carrying 4× fewer parameters—supporting H1 and exposing a distillation paradox we return to in Sec. 9. The soft teacher targets evidently regularise the student rather than merely compress it. Per-class (Tab. 5, Fig. 4), the model is near-perfect on Superficial-Intermediate and Parabasal (F_1 0.995/0.994) and weakest on Koilocytotic (F_1 0.919, recall 0.873), whose koilocytic halo overlaps morphologically with neighbouring classes; this single class accounts for most of the in-distribution error.

Table 4. In-distribution five-class metrics. SIPaKMeD validation, n=1003. The distilled student surpasses its teacher on every metric and is well calibrated.

Metric	Teacher (B4)	Student (B0)
Parameters	18.5 M	4.67 M
Accuracy	0.943	0.966
Balanced accuracy	0.946	0.968
Macro-F_1	0.945	0.967
Cohen's κ	0.929	0.958
AUROC (OvR macro)	0.995	0.998
ECE (5-class)	0.013	0.015

7.3. Calibration and explainability (answering RQ4, in-distribution)

Both models are well calibrated in-distribution: ECE 0.013 (teacher) and 0.015 (student) on the five-class problem, falling to 0.008 for the student on the binary Normal/Abnormal decision, with reliability bins close to the identity (Fig. 5). We carry the binary value forward as the in-distribution calibration reference, because the cross-dataset evaluation of Sec. 8 is

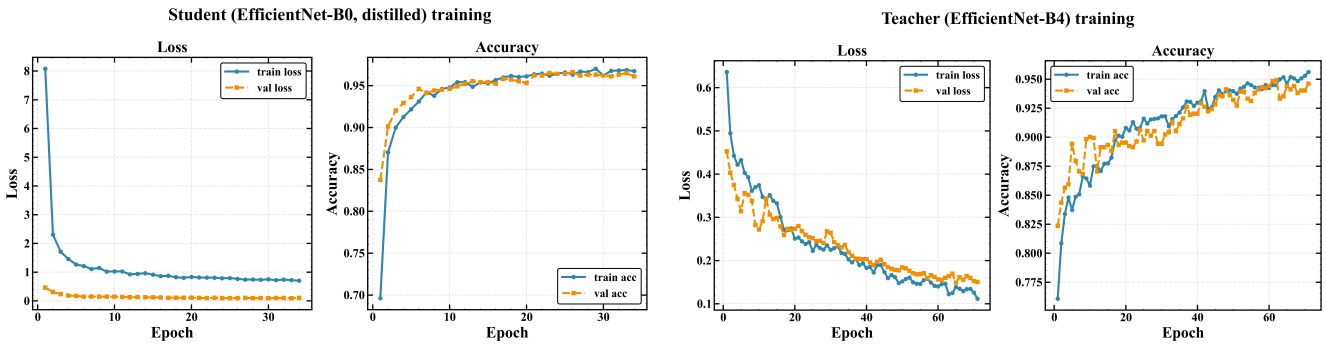


Figure 3. Training dynamics. Left: distilled B0 student (34 epochs). Right: B4 teacher (two stages, 71 epochs). Validation tracks training in both—no overfitting.

Table 5. Per-class metrics (student). Precision, recall and F_1 on the SIPaKMeD validation split. Koilocytotic is the weakest class.

Class	P	R	F_1	n
Superficial-Intermediate	0.990	1.000	0.995	191
Parabasal	0.989	1.000	0.994	179
Dyskeratotic	0.958	0.986	0.971	207
Metaplastic	0.933	0.981	0.957	213
Koilocytotic	0.969	0.873	0.919	213

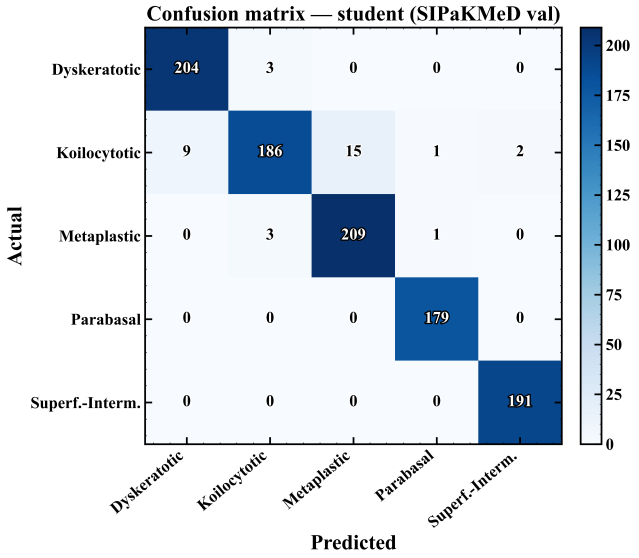


Figure 4. Five-class confusion matrix (student). Errors concentrate on Koilocytotic.

binary. Grad-CAM [10] over the student's final convolutional block (Fig. 6) localises the evidence to the nucleus and cell body for one representative correctly-classified cell per class (confidences $0.97-1.00$). The attributions are cytologically plausible—decisions are

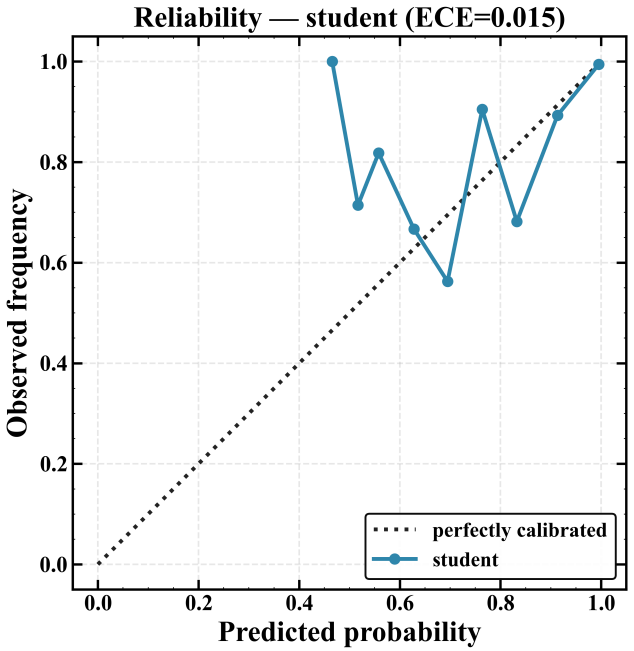


Figure 5. Reliability diagram (student). ECE 0.015 (five-class): in-distribution confidence is well calibrated and sits close to the identity.

driven by cell structure, not background artefacts—supporting the in-distribution half of H4. We stress the standard faithfulness caveat: Grad-CAM shows where the model looked, not that its reasoning is clinically correct. As Sec. 8 shows, this in-distribution calibration does not survive domain shift.

8. Cross-dataset collapse and its diagnosis

8.1. The collapse (answering RQ2)

Evaluated unchanged on the three external sets under the binary alignment of Tab. 3, the student's accuracy falls from 0.966 internally to 0.639 (Herlev), 0.605

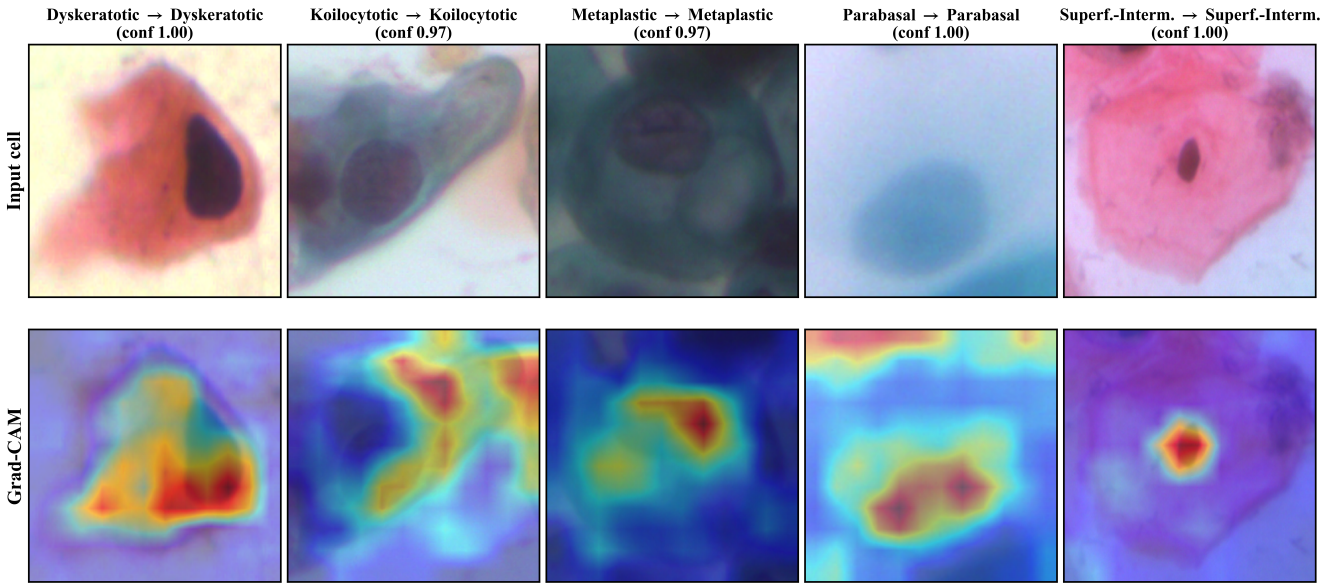


Figure 6. Grad-CAM attributions (student). One representative cell per class: top row, input cell; bottom row, Grad-CAM overlay (true → predicted, confidence). Heatmaps localise to nuclear/cellular regions; all five sample cells are correctly classified at confidence $0.97-1.00$.

(HiCervix) and 0.532 (APACC)—a 33-to-43-point drop (Herlev 33, HiCervix 36, APACC 43) that supports H2 (Tab. 6, Fig. 7). Threshold-free AUROC tells the same story more cleanly: 0.728, 0.594, 0.525. Two observations sharpen the result. First, on Herlev and HiCervix the accuracy at τ=0.5 is below the majority-class baseline ($0.639<0.74$; $0.605<0.621$, the latter only a razor-thin 1.6-point gap): a model that always predicted the majority class would score at least as well. Second, the error structure differs by dataset—the model over-predicts abnormal on APACC (specificity 0.417) and normal on HiCervix (sensitivity 0.386, missing most abnormal cells). A single global threshold is clearly mis-placed for each domain.

8.2. It is not a labelling artefact (answering RQ3)

The decisive question is whether this collapse is an artefact of our binary crosswalk. It is not. If the mapping were flipped or mis-defined, the decision score $P(\text{abn})$ would rank abnormal cells below normal ones and AUROC would fall below 0.5. The opposite holds in every dataset: the class-conditional separation $\Delta P = \overline{P(\text{abn})}_{\text{abn}} - \overline{P(\text{abn})}_{\text{norm}}$ is positive everywhere (+0.284 Herlev, +0.112 HiCervix, +0.045 APACC; Fig. 8, Tab. 6) and every AUROC exceeds chance. The score is correctly oriented—the model knows, weakly, which cells are abnormal—so the labels are not the problem. We additionally audited each per-dataset mapper against the native taxonomies (Tab. 3); the stricter Herlev-columnar variant shifts the headline by <1 point. H3 is supported: the failure is genuine, not an alignment artefact.

8.3. What actually breaks

Three mechanisms, jointly, explain the numbers.
(1) Operating-point miscalibration that does not transfer. The in-distribution confidence is well calibrated (binary ECE 0.008), but this property is domain-local: external binary ECE balloons to 0.256 (Herlev), 0.219 (HiCervix) and 0.328 (APACC)—a $27-41\times$ increase over the internal value (Tab. 6). The $P(\text{abn})$ distribution shifts wholesale per domain—mean $P(\text{abn})$ is 0.61 on APACC (vs. a base abnormal rate of 0.49, hence over-calling abnormal) and 0.35 on HiCervix (vs. 0.38, hence under-calling)—so the fixed τ=0.5 lands in the wrong place and discards the ranking signal that AUROC shows is present. This is exactly why accuracy can fall below the majority baseline while AUROC stays above chance. Quantifying the gap, a per-dataset oracle threshold (chosen on the test set, reported only to bound the calibration loss) recovers accuracy to 0.75 (Herlev, τ=0.05) and 0.65 (HiCervix, τ=0.85)—but not APACC (0.53), which has no signal to recover. The non-transfer of calibration is the cross-dataset half of H4.
(2) Signal degrades with slide-preparation distance. The retained AUROC is ordered by how far each dataset's preparation sits from SIPaKMeD's conventional smear: Herlev (conventional) 0.73 > HiCervix (liquid-based) 0.59 > APACC (auto-segmented noisy

Table 6. Cross-dataset binary results (student). Internal SIPaKMeD accuracy is 0.966. ECE is the binary expected calibration error (15 bins; in-distribution binary ECE is 0.008); ΔP is the separation between mean $P(\text{abn})$ on true-abnormal vs. true-normal cells (positive = correct ordering); "Maj." is the majority-class baseline. At τ=0.5, accuracy falls below the majority baseline for Herlev and HiCervix despite AUROC$>$ 0.5, and ECE is 27$-$41$\times$ the internal value.

Dataset	n	Acc	Bal. acc	Sens	Spec	AUROC	ECE	ΔP	Maj.
Herlev	917	0.639	0.651	0.625	0.678	0.728	0.256	+0.284	0.74
HiCervix	4823	0.605	0.562	0.386	0.738	0.594	0.219	+0.112	0.62
APACC	1125	0.532	0.535	0.654	0.417	0.525	0.328	+0.045	0.51

Internal SIPaKMeD: Acc 0.966, binary ECE 0.008. Higher ECE = worse calibration.

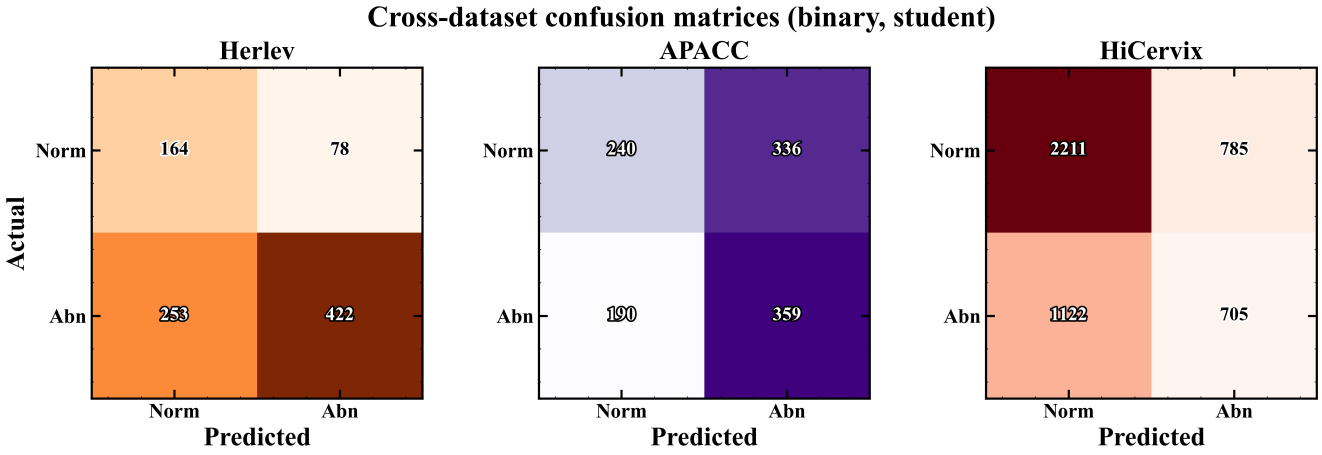


Figure 7. Cross-dataset confusion matrices (binary, student). The model over-predicts abnormal on APACC (low specificity) and normal on HiCervix (low sensitivity).

crops) 0.53$\approx$chance. The model has, in effect, learned features entangled with the preparation pipeline—a cytology instance of shortcut learning [3]—and the discriminative signal erodes monotonically as that pipeline diverges.

(3) The internal number is itself optimistic. The 0.966 is measured on a random per-cell split, which places cells from the same field-of-view into both train and validation; those cells share staining, focus and debris, so the validation estimate is inflated by leakage.

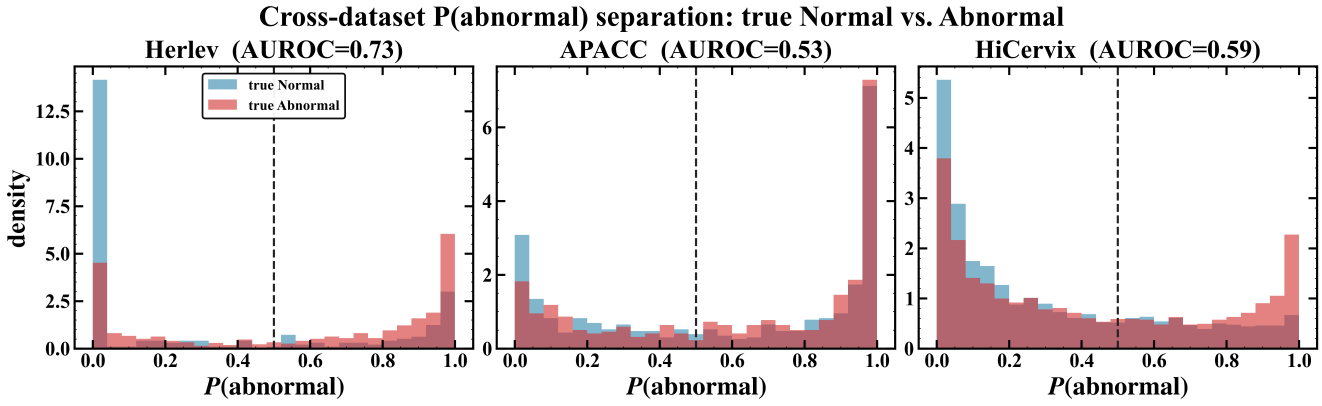


Figure 8. $P(\text{abn})$ separation. Distribution of the decision score for true-normal (blue) vs. true-abnormal (red) cells. Separation is clear on Herlev (AUROC 0.73), weak on HiCervix (0.59) and essentially absent on APACC (0.53). The dashed line is τ=0.5; a flipped label map would swap the colours and push AUROC below 0.5.

A leakage-safe, cluster-grouped split (no field-of-view shared across partitions) would lower it, widening the true internal-to-external gap beyond the 33−43 points measured here. The honest headline is therefore worse than even Tab. 6 suggests.

9. Discussion

The distillation paradox. The student is smaller and stronger than its teacher in-distribution (0.966 vs. 0.943, every metric; Tab. 4) yet no more robust out-of-distribution (both collapse under the same domain shift). Distillation here acts as a regulariser—the temperature-softened teacher targets smooth the student's decision surface on the training distribution—but it transfers the teacher's inductive biases, including its reliance on preparation-entangled features. Compression bought accuracy and efficiency, not generalisation. A practitioner who reads only the in-distribution number would deploy a model that is wrong more often than a constant predictor on two of three external sites.

Why a single in-distribution metric misleads. Three numbers that look reassuring in isolation each hide the failure: accuracy (0.966) is inflated by field-of-view leakage; ECE (binary 0.008 in-distribution) is domain-local and balloons to 0.22–0.33 under shift; and AUROC stays > 0.5 externally even where accuracy is below the majority baseline, so a threshold-free metric alone would also mislead in the opposite direction. The collapse is only visible when these are read together and per-domain—which is the audit protocol we advocate.

What would actually help. Our diagnosis points to specific, testable remedies rather than generic "train on more data". (i) Per-domain threshold calibration: the oracle analysis shows a single re-tuned τ recovers 0.75/0.65 on Herlev/HiCervix, so even a handful of labelled target cells would substantially help where ranking signal survives. (ii) Preparation-invariant features: stain normalisation [7] and preparation-aware augmentation to break the shortcut [3]. (iii) Leakage-safe evaluation: cluster-grouped splits so the internal number is honest from the start. (iv) Report the floor: always print the majority baseline next to accuracy, so a below-baseline operating point cannot hide.

Limitations. We audit one architecture family (EfficientNet) and one training source (SIPaKMeD); the magnitude of collapse may differ for other backbones, though the mechanism—non-transferable operating point plus preparation shortcut—is general. The oracle thresholds are computed on the test set and are an upper bound on what calibration could recover, not a deployable result. The binary crosswalk, while deterministic and documented, discards glandular and out-of-scope cells; we report that excluded fraction but do not evaluate on them. Grad-CAM is a localisation, not a faithfulness, guarantee. None of these qualifies the central finding: strong internal numbers do not imply cross-site readiness, and the gap is measurable and mechanistically explainable.

10. Conclusion

We audited a knowledge-distilled cervical-cytology classifier the way it would be used, not the way it is usually reported: we froze it after strong in-distribution training and measured it on three external Pap datasets spanning the slide-preparation axis. The result is a clean cautionary tale. In-distribution the distilled B0 student is excellent and beats its own B4 teacher (accuracy 0.966, AUROC 0.998, ECE 0.015) at a quarter of the parameters. Pushed across datasets, the same weights collapse to 0.639/0.605/0.532 accuracy and 0.728/0.594/0.525 AUROC—at two sites below what a constant majority predictor would score.

Our central contribution is the diagnosis, not the collapse. We show it is not a labelling artefact—the $P(\text{abn})$ score is correctly oriented everywhere ($\Delta P > 0$, AUROC> 0.5)—but the joint product of an operating point that does not transfer (in-distribution calibration is domain-local: binary ECE rises from 0.008 to 0.22–0.33 across sites) and features entangled with slide preparation (retained signal falls monotonically with preparation distance: conventional $>$ liquid-based $>$ auto-segmented). We further show the internal 0.966 is itself optimistic under field-of-view leakage, so the true gap is wider still. Each finding is tied to a falsifiable test and a concrete remedy—per-domain threshold calibration, preparation-invariant features, leakage-safe splits, and always reporting the majority floor.

The broader lesson generalises past cytology: a single in-distribution metric—accuracy, calibration, or AUROC alone—can each independently hide a model that is unfit for cross-site deployment. Honest medical-imaging evaluation must be multi-metric, per-domain, leakage-aware, and reported against the majority baseline. We release the full audit protocol and per-dataset breakdowns (Appendix A) to make such reporting routine.

A. Reproducibility and protocol details

Environment. PyTorch 2.11 (CUDA 12.8), single NVIDIA RTX 5060 Ti (16 GB), BF16 autocast, deterministic seed 42, num_workers= 0. All numbers in this paper are produced by the frozen checkpoints; we report measured values only and do not extrapolate.

Models and optimisation. EfficientNet-B4 teacher

(18.5 M) and B0 student (4.67 M), ImageNet-pretrained, custom head BN → Linear(512) → ReLU → Dropout → Linear(5). Teacher: Adam, class-weighted CE, two stages (frozen lr 10^{-3} ≤30 ep; fine-tune lr 10^{-5} ≤ 50 ep), ReduceLROnPlateau on val loss. Student: KD (Eq. (2), T=4, α=0.1), Adam lr 5×10^{-5}, ≤40 ep, early-stopped on val accuracy. Input 380×380, ImageNet normalisation; train augmentation = flips + rotation ±72° + affine scale 0.9−1.1; eval = resize + normalise only.

Split. Per-class stratified 80/20 over the 5,015 SIPaKMeD images (4,012 train / 1,003 val). We note explicitly (Sec. 8) that this random per-cell split shares field-of-view across partitions and therefore overstates internal accuracy; a cluster-grouped split is the recommended fix.

Binary decision and metrics. $P(\text{abn}) = \text{softmax}(z)_{\text{Koilo}} + \text{softmax}(z)_{\text{Dysk}}$ (Eq. (1)); predict Abnormal if $P(\text{abn}) > \tau$, τ=0.5. We report accuracy, balanced accuracy, sensitivity, specificity, threshold-free AUROC, the class-conditional separation ΔP, ECE (15 equal-width bins), and—next to every external accuracy—the majority-class baseline. Oracle thresholds are swept on the test set and reported only to bound recoverable calibration loss; they are not a deployable result.

External evaluation counts. Herlev 917 cells; APACC 1,125 crops (one zero-byte file removed; rubbish/bothcells excluded); HiCervix 4,823 scorable of 8,051 test cells (AGC/microbe excluded); Herlev columnar mapped to Normal in the legacy crosswalk (stricter exclusion variant moves the headline < 1 point). Excluded fractions are reported so that dropping unmappable cells cannot silently inflate the binary score.

Explainability. Grad-CAM [10] on the student's final convolutional block; one representative correctly-classified cell per class (Fig. 6, confidences 0.97−1.00). Attributions are localisation evidence, not a faithfulness guarantee.

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